

Supply, Demand, Government Policies, and the Efficiency of Markets

EC107: Microeconomics — policy & welfare

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Day 4

Price controls, surplus & efficiency, taxes, revenue & deadweight loss

Where we are going today

- ① A puzzle to open
- ② Controls on prices
- ③ Welfare economics: is the outcome good?
- ④ Consumer surplus
- ⑤ Producer surplus
- ⑥ Market efficiency
- ⑦ Taxes: the wedge and who pays
- ⑧ Government revenue
- ⑨ Deadweight loss
- ⑩ Incidence and elasticity
- ⑪ Efficiency, equity, and market failure
- ⑫ Wrap-up: pitfalls & takeaways

Two levers — and a market that overrules them

A government has two blunt instruments for bending a market:

- ① **Set the price** directly — a ceiling or a floor.
- ② **Tax the transaction** — and choose who legally hands over the money.

Today's claims, which sound wrong

- Capping a price to help buyers can leave them *worse off*.
- It makes **no difference** whether a tax is legally placed on the buyer or the seller.
- Every tax destroys value that goes to *nobody* — and that loss grows with the *square* of the tax rate.

We will prove each — and then ask the deeper question: left alone, does the market already do something no planner could? Tools: supply, demand, elasticity, and **surplus**.

Two kinds of price control

Price ceiling

A legal **maximum** price.

Binds only if set **below** equilibrium.

Example: *rent control*.

Price floor

A legal **minimum** price.

Binds only if set **above** equilibrium.

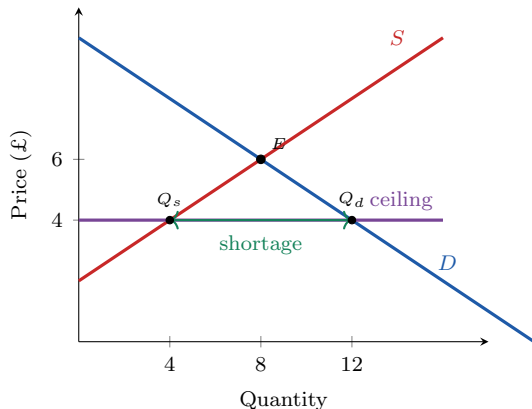
Example: the *minimum wage*.

A control on the “wrong” side of equilibrium is **non-binding** — the market never reaches it, so it does nothing. The interesting (and damaging) cases are the binding ones.

Key idea

Price is how a market *rations* a good. Freeze the price and you don't remove the scarcity — you just force *another*, worse, rationing device to appear.

A binding price ceiling causes a shortage

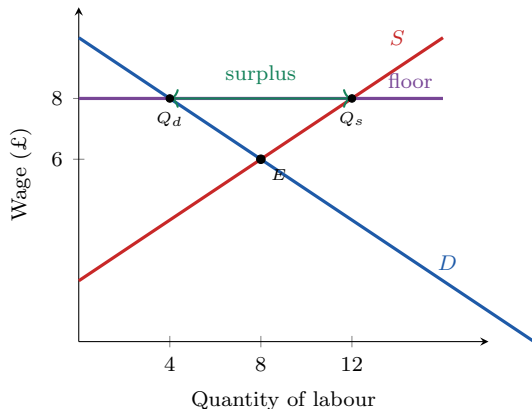


Ceiling at £4 ($< £6$) binds: $Q_s = 4$, $Q_d = 12$, **shortage** = 8.

Rent control's hidden costs

Cheaper rent for the lucky insiders who hold a flat — but for everyone else: queues, waiting lists, “key money” and bribes, discrimination, and landlords letting quality rot. The price can no longer ration, so *non-price* rationing takes over — usually less fairly.

A binding price floor causes a surplus



Floor at £8 ($> £6$) binds: $Q_d = 4$, $Q_s = 12$, **surplus** = 8.

The minimum wage

A labour surplus *is* unemployment: those who keep jobs earn more, but fewer are hired, and the burden lands on the low-skilled and young. *How large* that effect is is genuinely contested empirically — monopsony power, and the level relative to local wages, both matter. The *direction* is what the model pins down.

Your turn — price controls

Your turn

The equilibrium price of bread is £2.

- ❶ The government sets a price ceiling of £3. What happens?
- ❷ It lowers the ceiling to £1. Now what?
- ❸ Name two ways the good might be rationed once price cannot do the job.

(1) Non-binding (above equilibrium) — no effect. (2) Binding — a shortage of bread. (3) E.g. queues / first-come-first-served, seller favouritism or discrimination, black markets, rationing coupons.

Smith's claim — faith or theorem?

We have just seen price controls force shortages and surpluses — but were those outcomes truly *bad*? To say so, we first need a way to **measure** “good”.

Adam Smith, 1776

Individuals pursuing only their own gain are led “as if by an **invisible hand**” to an outcome good for society as a whole. That is a *testable* claim — but only once we can measure “good”.

So far we know *what* a market produces; now we ask whether it produces the *right amount*. Our measuring stick is **total surplus** — the value the market creates, shared between buyers and sellers. We will reach a sharp result:

The competitive equilibrium maximises total surplus. Smith's hand is a theorem — under assumptions we will state honestly.

Willingness to pay and consumer surplus

Definitions

Willingness to pay (WTP): the most a buyer will pay — their private *valuation*.

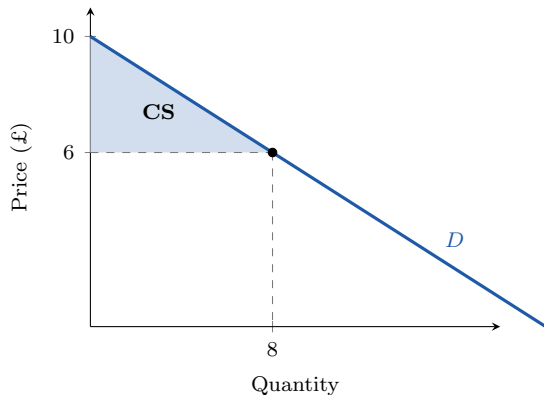
Consumer surplus = WTP – price paid: the “bargain” the buyer walks away with.

Imagine four buyers for one concert ticket, valuing it at £100, £80, £60, £40. At a price of £55:

- Three buy (values 100, 80, 60 > 55); surplus = 45 + 25 + 5 = £75.
- The £40 buyer walks away — forcing her would *destroy* value.

Stack enough buyers and the steps smooth into the **demand curve**: its height at each quantity is the marginal buyer's WTP, so **CS is the area below demand and above the price**.

Consumer surplus is a triangle



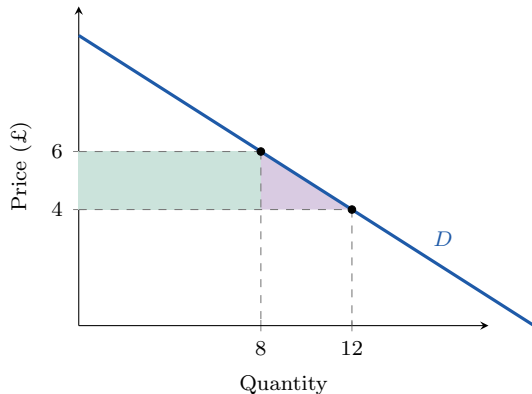
At £6 and $Q = 8$:

$$CS = \frac{1}{2} \times 8 \times (10 - 6) = \text{£}16.$$

- Base = quantity = 8.
- Height = choke price – price = 4.

Every buyer to the left of 8 valued the good above £6; the shaded area is their collective gain.

A price fall raises CS through two channels



Price falls £6→£4:

- **Existing** buyers (0–8) pay £2 less each: +£16 (green box).
- **New** buyers (8–12) enter: +£4 (purple triangle).

Total CS rises £16→£36. The same two effects, in reverse, are why a price *rise* hurts consumers.

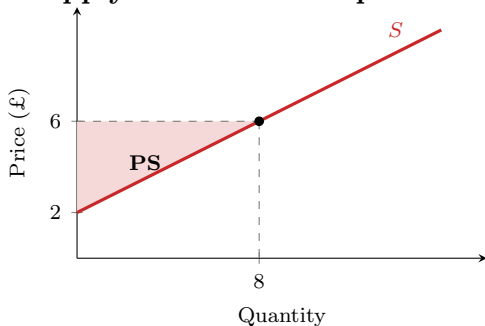
Cost and producer surplus

Definitions

Cost: the lowest price a seller will accept — the value of resources used, i.e. *marginal cost*.

Producer surplus = price received – cost.

The supply curve's height at each quantity is the marginal seller's cost, so **PS is the area above supply and below the price**.

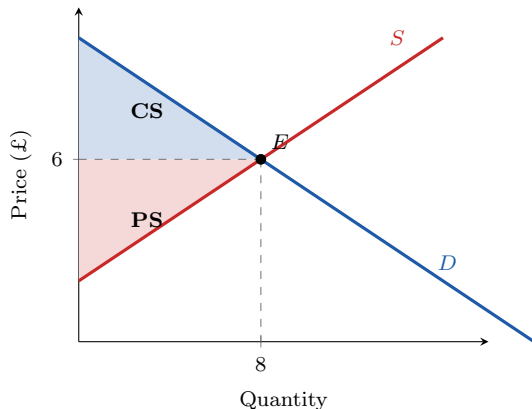


At £6, $Q = 8$:

$$PS = \frac{1}{2} \times 8 \times (6 - 2) = \text{£}16.$$

Sellers whose costs lay below £6 pocket the difference. (*Strictly, supply is the marginal-cost curve above the shut-down point — the firm theory comes later.*)

Total surplus, and why equilibrium maximises it



$TS = CS + PS = 16 + 16 = £32$ — the whole area between demand and supply up to Q^* .

- **Below Q^* :** value $>$ cost — an untraded unit would add surplus.
- **Above Q^* :** cost $>$ value — producing it would destroy surplus.

Only at Q^* is every gainful trade made and no harmful one. **Equilibrium maximises total surplus.**

The three efficiency properties of a free market

- ① The good is consumed by the buyers who **value it most** (they outbid the rest).
- ② The good is produced by the sellers who can make it **most cheaply** (they undercut the rest).
- ③ The **quantity** is exactly the one that maximises total surplus.

The invisible hand, stated precisely

No planner sets prices, matches buyers to sellers, or chooses the quantity. Yet self-interest plus a free price delivers the surplus-maximising allocation. *Prices carry the information and the incentives that a planner would need — and could never assemble.*

Efficiency here means *no reallocation can make someone better off without making someone worse off.*

Your turn

Our market: $D : P = 10 - \frac{1}{2}Q$, $S : P = 2 + \frac{1}{2}Q$, equilibrium $(8, 6)$.

- 1 Compute CS, PS and total surplus at equilibrium.
- 2 A regulator forces output up to $Q = 10$ (price held so it sells). Does total surplus rise or fall? Why?
- 3 In one sentence: what does the demand height *below* the supply height (for units 8–10) represent?

(1) $CS = 16$, $PS = 16$, $TS = 32$. (2) Falls — units 8–10 cost more to make than buyers value them, destroying surplus (over-production). (3) The amount by which cost exceeds value on each over-produced unit — surplus destroyed.

A tax drives a wedge between two prices

The competitive market just gave us the efficient benchmark. Now we watch a tax pull the outcome *away* from it — starting with the question of who really bears the cost.

Tax incidence

The study of *who actually bears* the burden of a tax — which need not be who legally pays it.

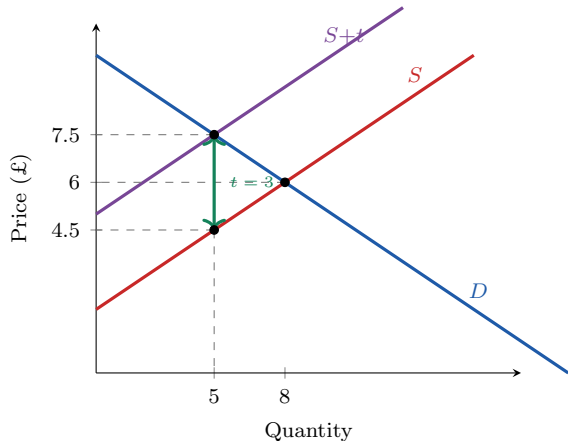
A per-unit tax of $\pounds t$ splits the single market price into *two*:

$$\underbrace{P_b}_{\text{price buyers pay}} - \underbrace{P_s}_{\text{price sellers keep}} = t.$$

Quantity falls until the gap between what buyers will pay and what sellers will accept is exactly t . **We will show the outcome is the same no matter which side the law taxes.**

First, tax the sellers.

A £3 tax on sellers

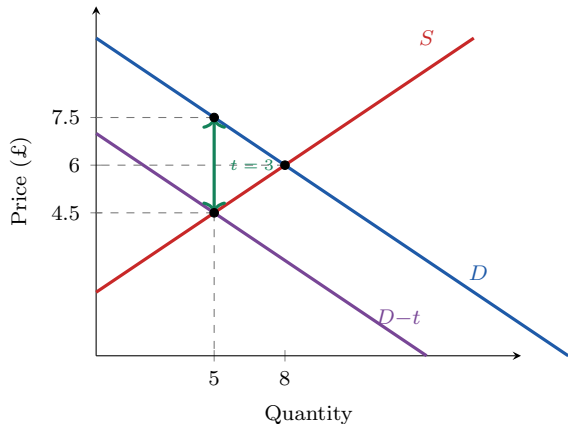


Sellers now need £3 more for every unit, so supply shifts **up** by 3 ($S \rightarrow S+t$).

- Quantity: $8 \rightarrow 5$.
- Buyers pay $P_b = £7.5$.
- Sellers keep $P_s = 7.5 - 3 = £4.5$.

Relative to the old £6: buyers +£1.5, sellers -£1.5. A 50/50 split *here* because the slopes are symmetric.

The same £3 tax *on buyers*



Buyers will now pay sellers £3 less (they owe it to the taxman), so demand shifts **down** by 3 ($D \rightarrow D-t$).

- Quantity: $8 \rightarrow 5$ — *the same*.
- Sellers keep $P_s = £4.5$; buyers pay $P_s + 3 = £7.5 = P_b$.

Identical quantity, identical P_b and P_s . The only thing that moved is the *posted* price — an accounting label.

Statutory side is irrelevant — the algebra

Let demand be $P = 10 - \frac{1}{2}Q$ and supply $P = 2 + \frac{1}{2}Q$, with a per-unit tax t .

Tax on sellers. They receive $P_s = P_b - t$ and supply on $P_s = 2 + \frac{1}{2}Q$:

$$P_b - t = 2 + \frac{1}{2}Q, \quad P_b = 10 - \frac{1}{2}Q.$$

Subtract: $10 - \frac{1}{2}Q - t = 2 + \frac{1}{2}Q \Rightarrow Q = 8 - t$.

Tax on buyers. They pay $P_b = P_s + t$ and demand on $P_b = 10 - \frac{1}{2}Q$:

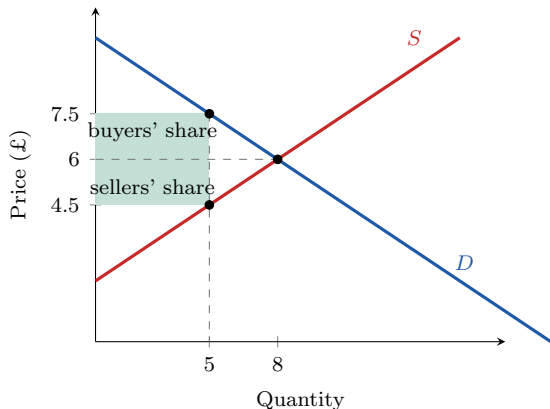
$$P_s + t = 10 - \frac{1}{2}Q, \quad P_s = 2 + \frac{1}{2}Q.$$

Subtract: $10 - \frac{1}{2}Q = 2 + \frac{1}{2}Q + t \Rightarrow Q = 8 - t$.

Same equations, same solution

Both give $Q = 8 - t$, hence the same P_b and P_s . **Who writes the cheque does not matter** — only *that* a wedge t exists. Real (economic) incidence is set by **elasticities**, not by the statute. “Tax the corporation, not the consumer” is political theatre.

Government revenue: the wedge $\times$ the quantity



Revenue = $t \times Q_{\text{tax}} = 3 \times 5 = \text{£}15$ — the shaded rectangle (height = t , width = Q).

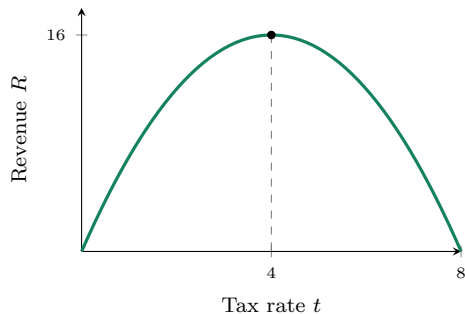
- Above the old £6: buyers' contribution = $(7.5 - 6) \times 5 = \text{£}7.5$.
- Below £6: sellers' contribution = $(6 - 4.5) \times 5 = \text{£}7.5$.

The £6 line splits the rectangle into who *really* funds the tax — here, equally.

Revenue is hump-shaped in the tax rate (the Laffer idea)

Revenue $R(t) = t \times Q(t)$, and $Q(t) = 8 - t$ falls as t rises. So:

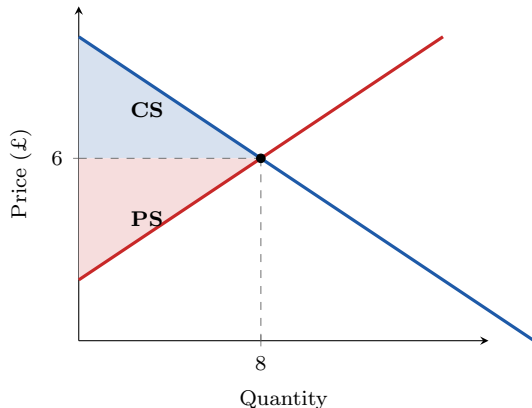
$$R(t) = t(8 - t) = 8t - t^2.$$



- Small t : revenue rises — the wedge dominates.
- Large t : revenue *falls* — the market has shrunk too far.
- Peak here at $t = 4$ (revenue £16); at $t = 8$ the market vanishes and revenue is *zero*.

A higher tax rate need not mean higher revenue — because it shrinks the base it taxes.

The welfare accounting: surplus before the tax

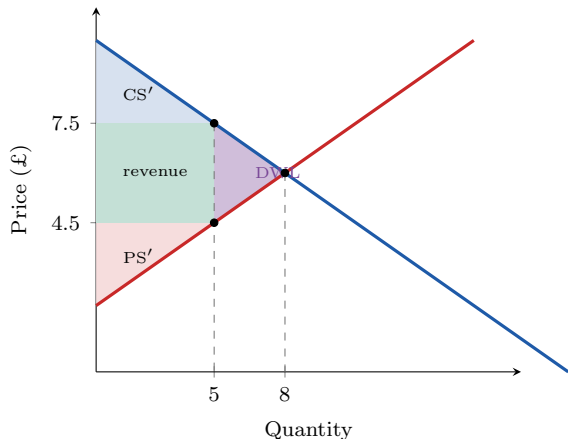


Recall our two measures of surplus:

- **Consumer surplus (CS):** area *below* demand, *above* price — value buyers get above what they pay.
- **Producer surplus (PS):** area *above* supply, *below* price.

At equilibrium: $CS = 16$, $PS = 16$, so **total surplus** = £32. This is the value the free market creates. Now watch the tax shrink it.

Where the surplus goes after the tax



	before	after
CS	16	6.25
PS	16	6.25
Gov. revenue	0	15
Total	32	27.5

Buyers and sellers lose
 $16 + 16 - 6.25 - 6.25 = £19.5$; the government collects £15. The missing **£4.5** is the **deadweight loss**.

Why deadweight loss exists

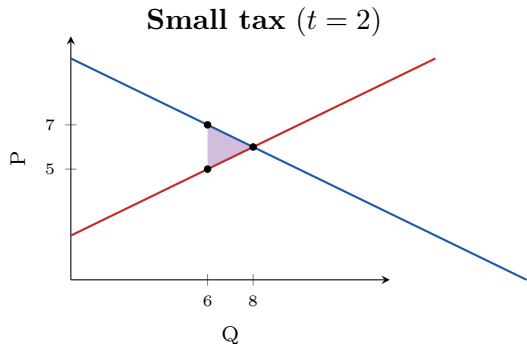
The tax shrinks quantity from 8 to 5. Every unit between 5 and 8 was one where the buyer's *value* exceeded the seller's *cost* — a mutually beneficial trade. The tax makes those trades not worth doing, and that surplus simply **vanishes**.

$$DWL = \frac{1}{2} \times \underbrace{(8 - 5)}_{\Delta Q} \times \underbrace{3}_t = \text{£}4.5.$$

The crucial distinction

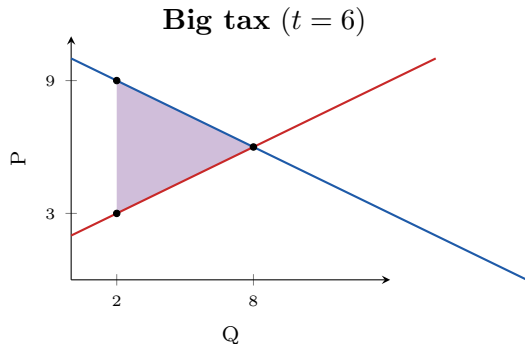
Government revenue is a *transfer* — value moved from buyers and sellers to the state, not destroyed. Deadweight loss is value **destroyed** — it reaches buyer, seller *and* state alike: nobody. It is the true economic cost of the tax, over and above the money raised.

What makes deadweight loss large?



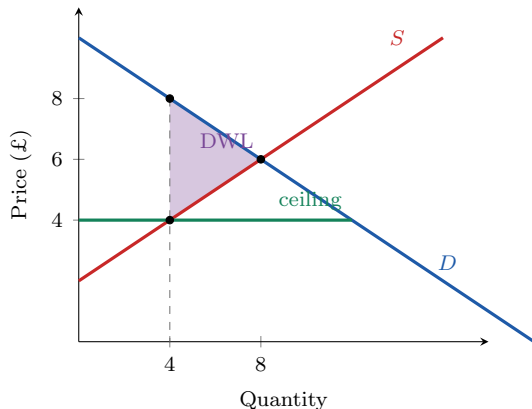
$$DWL = \frac{1}{2} \cdot 2 \cdot 2 = \pounds 2$$

Here $DWL = \frac{1}{2}t^2$: **tripling** the tax ($2 \rightarrow 6$) raises DWL **nine-fold** ($2 \rightarrow 18$). DWL grows with the *square* of the rate, and is larger the more **elastic** either curve is.



$$DWL = \frac{1}{2} \cdot 6 \cdot 6 = \pounds 18$$

Deadweight loss from a price ceiling



A ceiling at £4 lets only $Q = 4$ be supplied. Units 4–8 — all with value above cost — go untraded.

$$DWL = \frac{1}{2} \times (8 - 4) \times (8 - 4) = £8.$$

Same disease, different cause

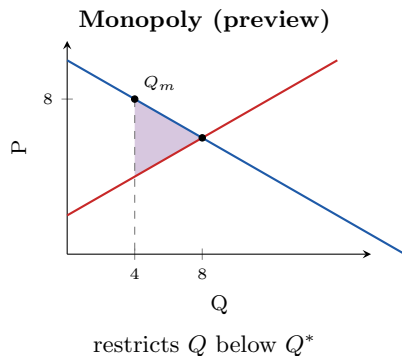
A tax did this with a wedge; a control does it with a frozen price. *Any* force that pushes quantity off Q^* destroys the surplus on the lost trades.

One idea, many guises

Deadweight loss is the triangle between *value* (demand) and *cost* (supply) over the units that **should** have traded but didn't (or shouldn't have but did).

Forces that open a wedge between value and cost:

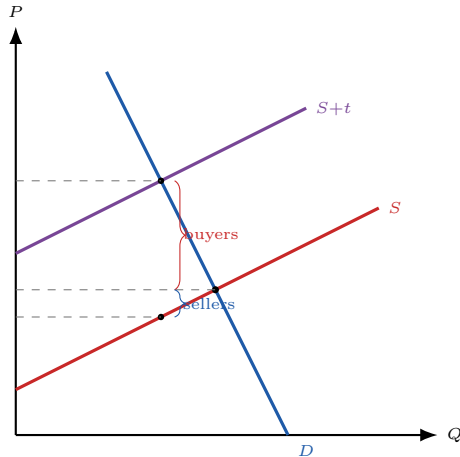
- **Taxes** — a wedge of size t .
- **Price controls** — ceilings and floors.
- **Market power** — a monopolist restricts output to lift price.
- **Externalities** — private and social value diverge (later).



Elasticity decides who bears the tax

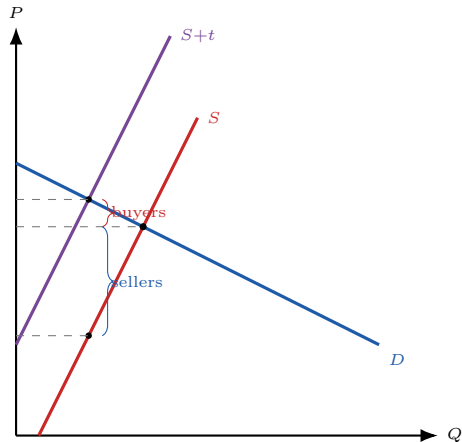
Inelastic demand

buyers bear most



Inelastic supply

sellers bear most



Rule: the burden falls on the **less elastic** side — the one less able to walk away.

Why, and where you see it

Intuition

Elasticity measures the ability to escape a bad price. The **inelastic** party has nowhere to go and absorbs the tax; the **elastic** party simply leaves, forcing the other to swallow it.

- **Cigarettes** (inelastic demand): the tax falls mainly on *smokers*.
- **Payroll tax**, statutorily split 50/50 between firm and worker: economic incidence falls mostly on *workers*, because labour *supply* is relatively inelastic — a vivid real example of statutory-side irrelevance.
- **US luxury-yacht tax (1990)** (elastic demand): buyers stopped buying / bought abroad; the burden hit *boat-builders*, and the tax was repealed.

Efficiency is not the same as fairness

Total surplus adds up everyone's gains *regardless of who gets them*. An allocation can be perfectly efficient and deeply unequal.

Two distinct questions

Efficiency: is the pie as large as possible? **Equity:** is the pie shared fairly?

- Economics has sharp tools for the first and far weaker ones for the second — equity involves value judgements that markets do not make.
- A policy that shrinks the pie (a deadweight loss) may still be chosen if it improves the *distribution* enough. Efficiency is one input to that decision, not the whole of it.

When the invisible hand misses: market failures

Our theorem rested on assumptions. Where they fail, equilibrium need *not* be efficient:

- **Market power** — a monopolist sets price above marginal cost and under-produces.
- **Externalities** — pollution, congestion: private cost $\neq$ social cost, so the market over- or under-produces.
- **Public goods & common resources** — non-excludable goods are under-supplied; shared resources are over-used.
- **Asymmetric information** — when buyers can't judge quality, good products can be driven out.

The honest summary

Free markets are usually a remarkably good default — the outcome to beat. “Usually” is not “always”, and identifying the genuine exceptions is most of the rest of microeconomics.

A case that pits efficiency against fairness

After a hurricane, the price of bottled water and generators spikes. Many places pass **anti-price-gouging** laws to stop it. Who is right?

The efficiency case *for* high prices

- Rations scarce supply to those who need it most.
- Signals sellers to rush more supply in.
- Avoids shortages, queues and empty shelves.

The fairness case *against*

- “Need most” becomes “can pay most” — the poor lose essentials.
- Exploits people in genuine distress.
- Windfalls feel unjust even if efficient.

This is the efficiency–equity tension made concrete: the analysis tells you the *cost* of the cap (a shortage, lost surplus); it cannot, by itself, tell you whether that cost is *worth* paying.

Misconceptions & exam traps — taxes & controls

- “Taxing the seller protects the buyer.” False — statutory side is irrelevant; elasticities decide.
- Confusing revenue with the cost of a tax. Revenue is a *transfer*; the social *cost* is the deadweight loss on top of it.
- Thinking a higher rate always raises more. Revenue is hump-shaped — the base shrinks.
- Forgetting the elasticity link. Bigger elasticities $\Rightarrow$ bigger $\Delta Q \Rightarrow$ bigger DWL, smaller revenue.
- Treating a binding control as harmless redistribution. It also destroys surplus and spawns non-price rationing.

Misconceptions & exam traps — efficiency & welfare

- “Efficient means fair.” No — total surplus ignores distribution entirely.
- Calling producer surplus “profit”. PS is revenue minus *variable* cost, not profit; fixed costs and the firm theory come later.
- Thinking more output is always better. Beyond Q^* , extra units cost more than they're worth — that destroys surplus too.
- Forgetting DWL is *destroyed*, not transferred. Tax revenue moves surplus; DWL erases it.
- Assuming markets are always efficient. The result holds *only* under its assumptions; market failures are the exceptions that prove the rule.

Key takeaways

- Binding **ceilings** $\Rightarrow$ shortages; binding **floors** $\Rightarrow$ surpluses; both force non-price rationing.
- **CS** = area below demand, above price; **PS** = area above supply, below price; **total surplus** is maximised at the competitive equilibrium — the invisible hand, proved.
- A tax opens a wedge $P_b - P_s = t$ and cuts quantity. **Who is taxed legally is irrelevant**; elasticities set the real burden (inelastic side pays more).
- **Revenue** $= t \times Q$ is a *transfer*, and is hump-shaped in the rate.
- **Deadweight loss** is value *destroyed* by any wedge between value and cost (tax, control, monopoly, externality); it grows with the *square* of the distortion.
- **Efficiency $\neq$ equity**, and the efficiency result holds only where markets don't fail.

The whole arc

One thread, start to finish

Ten principles → gains from trade → supply and demand → elasticity → government policy → the efficiency of markets. You now have the toolkit the rest of economics is built on.

Questions?

EC107: Microeconomics